Day 8 — JavaScript Fundamentals

Simple Questions, Answers, Examples, and Practice Programs

1. What is JavaScript?

JavaScript is a programming language used to make webpages interactive. It can respond to clicks, validate forms, update page content, and run in web browsers. It is also used on servers with environments such as Node.js.

2. What is the difference between var, let, and const?

var is function-scoped and can be redeclared. let is block-scoped and can be reassigned. const is block-scoped and cannot be reassigned after initialization. With const objects and arrays, their contents can still be changed.

```
var oldWay = 10;
let age = 20;
const country = "India";

age = 21; // allowed
// country = "Japan"; // TypeError
```

3. What are JavaScript data types?

Common primitive types are string, number, bigint, boolean, undefined, symbol, and null. Objects, including arrays and functions, are non-primitive values.

```
let name = "Asha";       // string
let age = 21;            // number
let passed = true;       // boolean
let result;              // undefined
let emptyValue = null;   // null
let skills = ["HTML", "JS"]; // array
let student = { id: 1, name: "Asha" }; // object
```

4. What are operators?

Operators perform calculations, comparisons, and logical checks. Examples: +, -, *, /, %, ===, !==, >, <, &&, ||, !, and =. Prefer === for comparison because it checks value and type without automatic type conversion.

```
let a = 10;
let b = 3;
console.log(a + b);  // 13
console.log(a % b);  // 1
console.log(a === 10); // true
```

5. What are conditional statements?

if, else if, and else allow a program to make decisions. switch is useful when comparing one expression against several cases.

```
let marks = 75;

if (marks >= 80) {
  console.log("Grade A");
} else if (marks >= 60) {
  console.log("Grade B");
} else {
  console.log("Needs improvement");
}
```

6. What are loops?

Loops repeat code. A for loop is common when the number of repetitions is known. while repeats while a condition is true. for...of iterates over values in an iterable such as an array.

```
for (let i = 1; i <= 5; i++) {
  console.log(i);
}

let numbers = [10, 20, 30];
for (const number of numbers) {
  console.log(number);
}
```

7. What are functions?

A function is a reusable block of code. It can accept parameters and return a result. Functions reduce repetition and make programs easier to organize.

```
function add(a, b) {
  return a + b;
}

console.log(add(5, 3)); // 8

const multiply = (a, b) => a * b;
console.log(multiply(4, 2)); // 8
```

8. What are arrays and common array operations?

An array stores an ordered collection of values. Common operations include push() to add an item, pop() to remove the last item, includes() to check for a value, and map(), filter(), and reduce() to process data.

```
const numbers = [3, 1, 2];
numbers.push(4);
```

```
console.log(numbers); // [3, 1, 2, 4]

const doubled = numbers.map(n => n * 2);
const greaterThanTwo = numbers.filter(n => n > 2);
console.log(doubled);
console.log(greaterThanTwo);
```

9. What are objects and how do you access their properties?

An object stores related information as key-value pairs. Use dot notation or bracket notation to access properties.

```
const student = {
  id: 101,
  name: "Asha",
  marks: 88
};

console.log(student.name);    // Asha
console.log(student["marks"]); // 88
student.course = "BSc";        // add property
```

10. Practice Program: Even or Odd

```
function evenOrOdd(n) {
  if (n % 2 === 0) {
    return "Even";
  }
  return "Odd";
}

console.log(evenOrOdd(7)); // Odd
```

11. Practice Program: Find the Largest of Three Numbers

```
function largest(a, b, c) {
  if (a >= b && a >= c) return a;
  if (b >= a && b >= c) return b;
  return c;
}

console.log(largest(12, 25, 9)); // 25
```

12. Practice Program: Factorial

```
function factorial(n) {
  if (!Number.isInteger(n) || n < 0) {
    return "Enter a non-negative integer";
  }

  let result = 1;
  for (let i = 2; i <= n; i++) {
    result *= i;
  }
  return result;
}

console.log(factorial(5)); // 120
```

13. Practice Program: Reverse a Number

```
function reverseNumber(n) {
  let reverse = 0;

  while (n > 0) {
    const digit = n % 10;
    reverse = reverse * 10 + digit;
    n = Math.floor(n / 10);
  }
  return reverse;
}

console.log(reverseNumber(1234)); // 4321
```

14. How can you run JavaScript?

You can place JavaScript inside a script element in an HTML file or put it in a separate .js file and load it using a script element. You can view console output using the browser's Developer Tools. Node.js can also run JavaScript files from a terminal.

```
<!-- In an HTML file -->
<script src="script.js"></script>
```

Day 8 — Task Checklist

- ■ Practise let, const, data types, and operators.
- ■ Write programs using if-else and loops.
- ■ Create functions for addition, subtraction, multiplication, and division.
- ■ Perform array operations using push, pop, map, and filter.
- ■ Create an object and read or update its properties.
- ■ Complete even/odd, largest of three, factorial, and number reversal programs.
- ■ Run code in a browser console or with Node.js.

Quick Revision

JavaScript: adds interactivity. **let:** reassigned block-scoped variable. **const:** block-scoped binding that cannot be reassigned. **Condition:** makes decisions. **Loop:** repeats code. **Function:** reusable logic. **Array:** ordered collection. **Object:** key-value data.